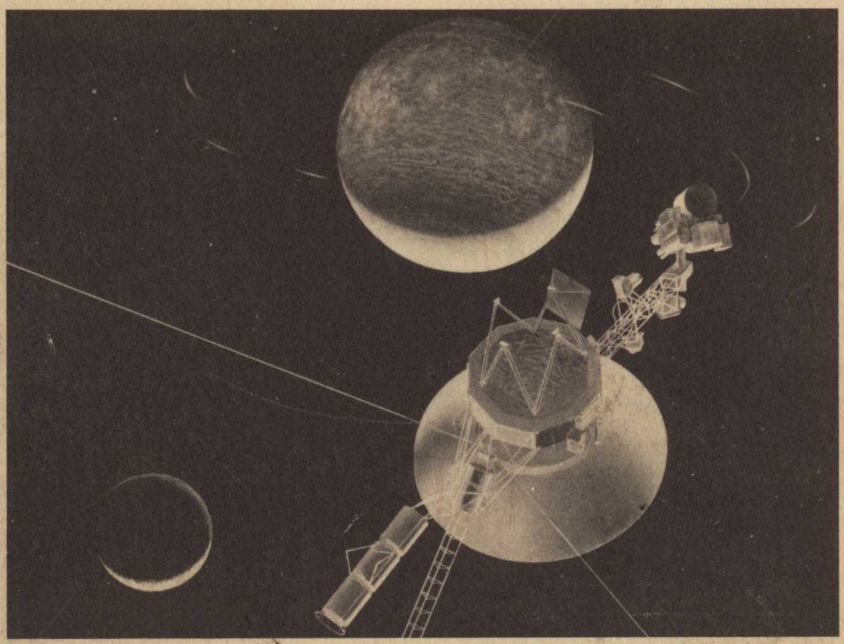
JPL Publication 89-24

The Voyager Neptune Travel Guide



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Man alone is the architect of his destiny.

William James

3. *GETTING THE JOB DONE*

When you are asked to think about space missions to the planets, you will probably think about stuff. Tangible stuff. You might think about the spacecraft, over 800 kilograms (nearly one ton) of structure and electronics gear. You might think of giant antennas to receive the signals from outer space. You might think of control centers with video displays and red, green, and yellow lights.

But you might not think of people, plans, and coordination. You should. This is the intangible stuff from which planetary missions are made. The theme of Voyager's people runs through this chapter of the Guide. Their performance through August 1989 will determine if Voyager Neptune earns a gold medal.

The gold medal, reminiscent of the XXIVth Olympiad held in Seoul in 1988, provides us a good comparison. The Voyager encounter with Neptune is an Olympian event, in terms of cost, complexity, number of people, and world-wide involvement.

Planning

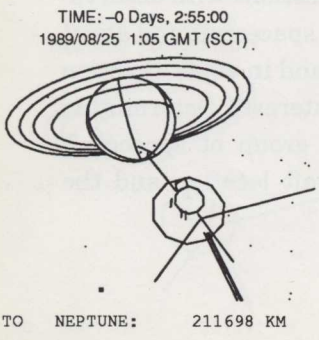
Start with the Voyager scientists. There are 130 Voyager investigators at universities, observatories, and aerospace companies scattered about the United States, Canada, England, France, West Germany, Italy, and the Soviet Union. The investigators are Voyager's competitors, by an elimination process at least as intense as the Olympic trials. They are supported by half again their number of research assistants and students. The investigators formulate the basic questions to be answered about Neptune, its satellites, and its neighborhood.

These questions require that Voyager 2 make specific observations at Neptune under just the right conditions. To match questions with observations requires a vast amount of information about the spacecraft and about Neptune. For example, just where is the spacecraft, and in what direction must the spacecraft look to see the various objects of interest? Determining the spacecraft position is the job of a 10-member group of spacecraft navigators at JPL who determine both the spacecraft location and the thruster firings needed to correct the location.



Figure 3-1. Charting Voyager's path through the solar system is a precise science, with only the occasional need to make artful choices among candidate "solutions."

Determining the locations of Triton and Nereid is initially the job of another 10-member group of orbit experts, some at JPL and some at places like the Universities of Texas and Virginia. They take the latest information available to refine the ephemeris (positions) of Neptune and its satellites. Remember, Neptune travels around the Sun so slowly that less than a Neptunian year has passed since its discovery, and less than three Neptunian months have passed since the discovery of its satellite Nereid by G. P. Kuiper in 1949. It's no wonder that there is still some uncertainty about precise locations, but these ephemeris experts can typically predict satellites positions to accuracies of 3000 km (1900 mi) or better many months in advance of the encounter. The predicted position of Nereid, however, has a much greater uncertainty.



The spacecraft navigators are after even better accuracies. They are equipped with some pretty fancy orbit determination computer programs that solve for the simultaneous positions of all bodies. Their most important data for updating the orbital elements of the satellites will be Voyager-2 images of the satellites against a known star background. By using the

narrow-angle TV camera to shoot satellite-star images until a few days before closest approach, these real-time navigators can estimate Triton's position relative to Neptune to 100 km (62 mi) or better. This is equivalent to locating the finish line for the marathon to an accuracy of 1/40 of an inch.

Okay, so we know where everything is. What's our master plan? We have limited resources and a rigid time schedule, but we want the greatest possible mission return. Voyager's Mission Planning Office (MPO) has the task of preparing a large collection of "guidelines and constraints" that govern how the Project resources and spacecraft consumables will be used to achieve high-value science return, while maintaining an acceptable level of risk. As suggested by Figure 3-2, the guidelines establish the envelope within which the mission sequences will be designed, implemented, and executed. The MPO function is analogous to that of the Olympic Organizing Committee. The time span is even comparable. Several of the basic decisions which were essential to the Neptune encounter were even made in the mid-1970s, well before launch.

Given knowledge of the satellite and target locations with time, scan platform pointing and observing conditions can be computed. The Science Investigation Support (SIS) Team at JPL, about 20 strong, acts as the focus

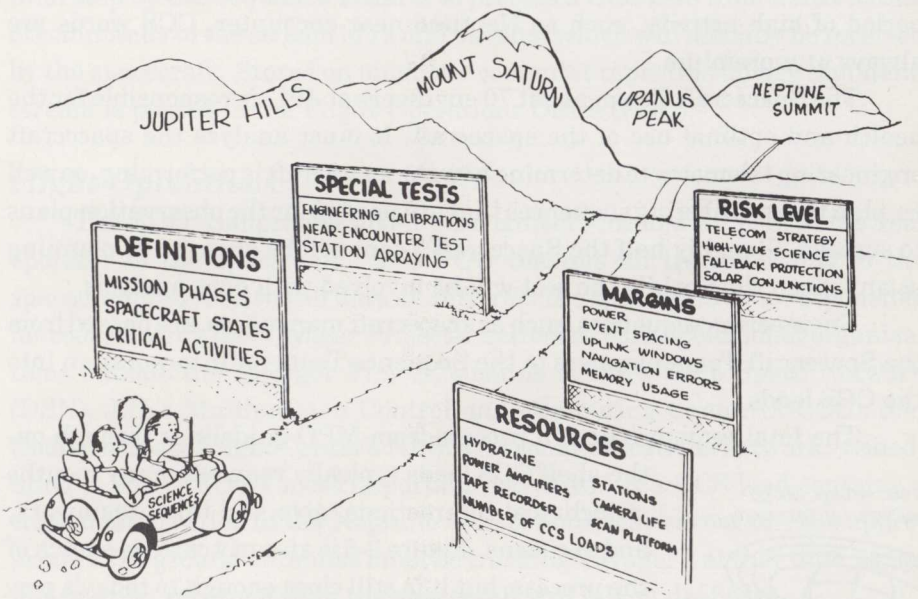


Figure 3-2. Mission planning establishes guidelines for the use of project consumables and helps define the envelope within which the sequences will be developed.

for the detailed science planning. Representatives for each experiment interact with investigators and other parts of the Voyager Project. The result is an integrated plan for all observations intended.

It is not uncommon for the investigators to desire conflicting observations. To resolve these conflicts, the Voyager Science Steering Group (SSG), composed of the Principal Investigators (the leaders of each of the eleven Voyager scientific teams), meets regularly at JPL to review the observation plans. The members of the SSG make the science decisions required to ensure that the observation plans will yield the best Neptune science.

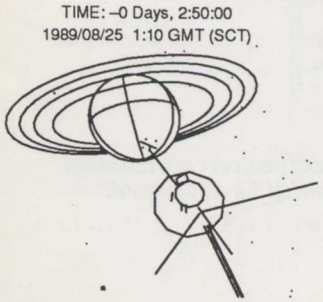
Sequencing

As soon as the final observation plan is ready, primary responsibility passes to the sequence development engineers. The Sequence Team consists of about 30 members at JPL who flesh out the observation plans with the instrument and spacecraft commands required to produce the desired results. The Sequence Team ensures that no operating constraints are violated and that all of the instructions to the spacecraft will fit into its computer memory. The Voyager-2 sequence computer, the Computer Command Subsystem (CCS), has roughly 2500 words of memory reserved for sequencing. Two words are required for a simple instruction: one to specify the event to take place, the other to specify the time of occurrence. For a period of high activity, such as Neptune near-encounter, CCS words are always at a premium.

The Spacecraft Team, about 70 engineers at JPL, is responsible for the health and optimal use of the spacecraft. It must analyze the spacecraft engineering telemetry to determine how the spacecraft is performing, as well as plan the engineering sequences that are needed for the observation plans to succeed. Roughly half the Spacecraft Team will be engaged in planning at any given time, while the rest will be involved with data analysis.

Engineering sequences, such as spacecraft maneuvers, are passed from the Spacecraft Team planners to the Sequence Team for incorporation into the CCS loads.

The total sequence design process from MPO guidelines through on-the-shelf CCS loads typically requires many months of technical interactions, give and take, teamwork, and decisions. Figure 3-3 is an early cartoon sketch of the process, but it is still close enough to today's perceptions to warrant inclusion in the Guide.



Once a CCS load has been built and verified, it is ready to be sent to the Voyager-2 spacecraft. The

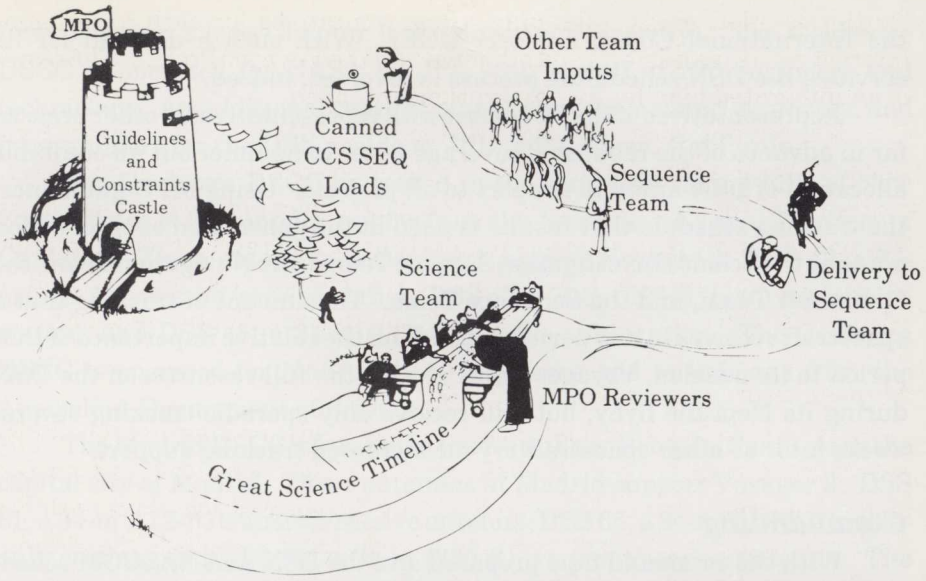


Figure 3-3. People often have humorous ways to view their working interrelationships, and Voyager is no exception (early version of "ye olde sequence design process").

final step by the Sequence Team is to process a CCS load from a text listing of commands to the stream of 1s and 0s (bits) which will actually be received by the spacecraft. Stored on magnetic computer tape, this binary command stream is passed to the Flight Operations Office (FOO).

Flight Operations

The FOO, comprising about 45 engineers located at JPL, is the real operator of the spacecraft. The FOO controls all transmissions to the spacecraft and receives all data at JPL from the spacecraft. It is responsible for coordinating the Voyager Project's activities with operational organizations outside the Voyager Project, such as NASA's Deep Space Network (DSN), JPL's Multimission Control and Computing Center (MCCC), and Goddard Space Flight Center's NASCOM communications network. Scheduling is one of FOO's most important activities. If a CCS load contains a critical observation of the Neptune atmosphere near encounter, the appropriate DSN ground antennas must be tracking Voyager 2 at that time, or the information returned from the spacecraft will fall on deaf ears.

During the Neptune encounter, the DSN will be tracking not only Voyagers 1 and 2, but also Pioneers 10, 11, and 12, and the Magellan probe to Venus. It may also be asked to track three earlier Pioneers (6, 7, and 8) and

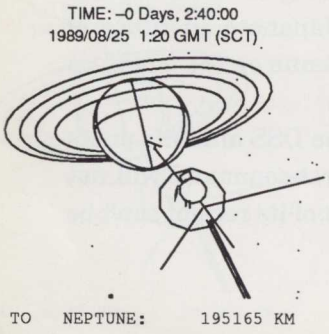


Figure 3-4. If we could move the 70-m DSN tracking antenna from its Goldstone, California, desert location to the football field inside the Pasadena Rose Bowl, this is how large it would appear! Big ears are needed at Earth Base to hear the feeble signals from a remote spacecraft.

seen at Earth until 8.2 hours after it was sent. Data telemetered from Voyager 2 are 4.1 hours old the instant they are received. This time lag complicates operations greatly. Imagine driving a car where the gauge readings and even the sights seen out of the windows are over four hours old, and the response to turning the steering wheel or applying the brake is four hours in the future! Luckily, there is less traffic on the way to Neptune than there is on the Los Angeles freeways.

Receiving Data

The data received are of enormous value. First there is telemetry, which consists of information describing the performance of the spacecraft and its instruments and the science measurements themselves. The telemetry is identified with spacecraft time so that a complete reconstruction of the state of the spacecraft can be used to determine the health of all engineering and science subsystems, and to aid in science interpretation.



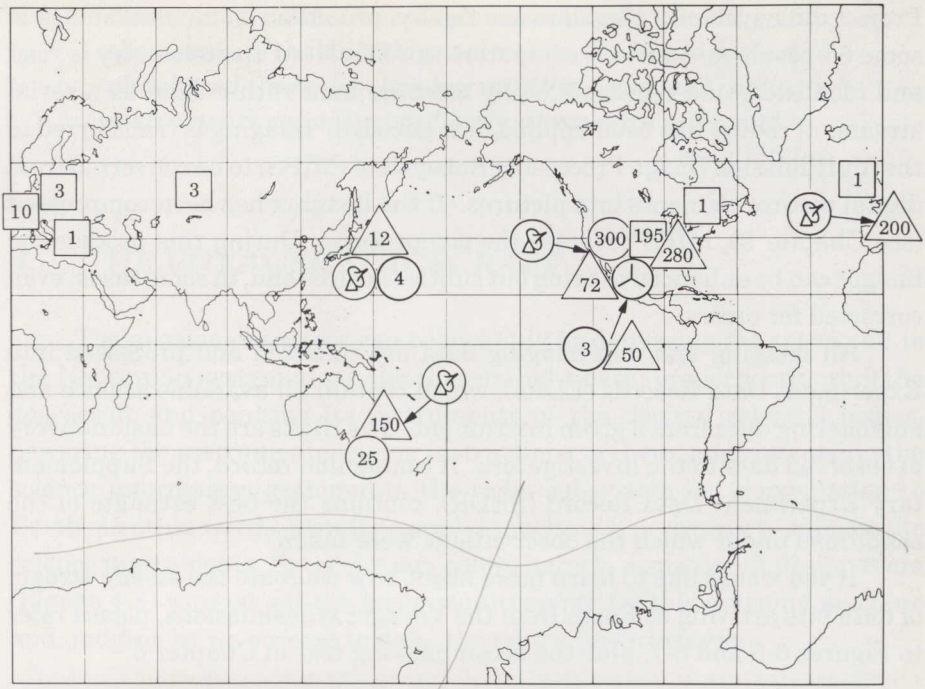


Figure 3-5. The Voyager family consists of full-time Voyager people (○), part-time Voyager people (□) and multimission support people (△). The tracking complex staffing levels shown represent the full complement of personnel who support not only Voyager, but several other missions as well.

Second there are navigation data. One type are doppler data, which are contained in the Voyager-2 radio signal itself and are dependent upon the relative motion between the spacecraft and tracking antenna. Another is range data, which provides a distance measurement from the spacecraft to the tracking antenna. A third uses simultaneous tracking by two stations of first the spacecraft and then a quasar of known characteristics to get a different type of doppler data. Finally, optical navigation video images relate the spacecraft and planetary body positions to their directions relative to known stars.

All of these data are both relayed (via GCF) to JPL and recorded at the DSCC. This allows any gaps to be recovered after the fact if any data are lost on the way to their ultimate user. The first users are the Mission Control Team, the Navigation Team, and the Spacecraft Team. Essential engineering measurements and status indicators are displayed as they are received so that corrective action may be initiated at any sign of a problem.

The entire data stream is routed through the Ground Data System (GDS), consisting of the DSN, NASCOM, MCCC and a number of Voyager

Project data systems. The data are finally processed by an assemblage of some 55 people and various computers at JPL. Here, the telemetry is read and identified, and reassembled by measurement rather than as a serial stream. If coding has been applied, it is decoded. Imaging is transferred to the Multimission Image Processing Subsystem (MIPS) to be converted from digital picture elements into pictures. If the imaging has been compressed (see Chapter 9), MIPS reverses the compression. During this process the images can be enhanced to bring out subtle features and, in some cases, even corrected for errors.

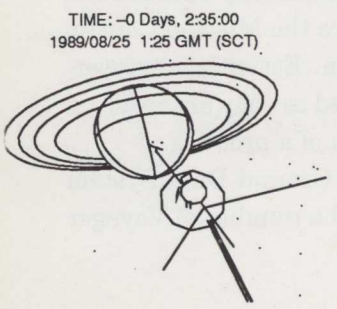
All imaging and non-imaging data are collected and processed into Experiment Data Records (EDRs), which contain all available science and engineering data from a given instrument. The EDRs are the basic delivery of observed data to the investigators. A companion record, the Supplementary Experiment Data Record (SEDR), contains the best estimate of the conditions under which the observations were taken.

If you would like to learn more about how we route the steady stream of data bits arriving at Earth from the Voyager transmissions, please refer to Figures 6-6 and 6-7, plus the accompanying text in Chapter 6.

The Results

As the encounter with Neptune approaches, delivery of the data to the investigators will become easier because the scientists will be migrating to JPL. As the pace quickens, monthly investigator meetings will become daily meetings, and ideas will be exchanged furiously as they strive to understand the details of a new planetary system. When the last of the Neptune data are safely acquired, the scientists will retreat with the data to their own institutions to begin the intensive process of converting measurements into answers to those fundamental questions raised in Chapter 2.

That won't be the end, however. Archived, the Voyager-2 Neptune data will be available to scientists the world over through the National Space Science Data Center at the NASA/Goddard Space Flight Center in Greenbelt, Maryland. It is a safe bet that somewhere today there are tens or hundreds of elementary school students who, in the early twenty-first century, will be writing doctoral dissertations based on their study of the Voyager-2 Neptune data. And that, in the year of the XXVIIth Olympiad, will be Voyager's real gold medal.



One wonders if their messages came long ago, hurtling into the swamp of the steaming coal forests, the bright projectile clambered over by hissing reptiles, and the delicate instruments running mindlessly down with no report.

Loren Eiseley

4. *THE VOYAGER SPACECRAFT*

The mission objectives can be met only by delivering the spacecraft to the Neptunian system along the chosen flight path, properly orienting the spacecraft and pointing its instruments at the desired celestial bodies, powering the instruments, giving instructions to them, and channeling the science information gathered to the radio subsystem for transmission to Earth. In other words, a pretty complex machine is necessary to support the science instruments. Several years before launch, a spacecraft design team (Figure 4-1) worked out the basic requirements for this amazing machine and, judging by its success to date, they did a first-class job.

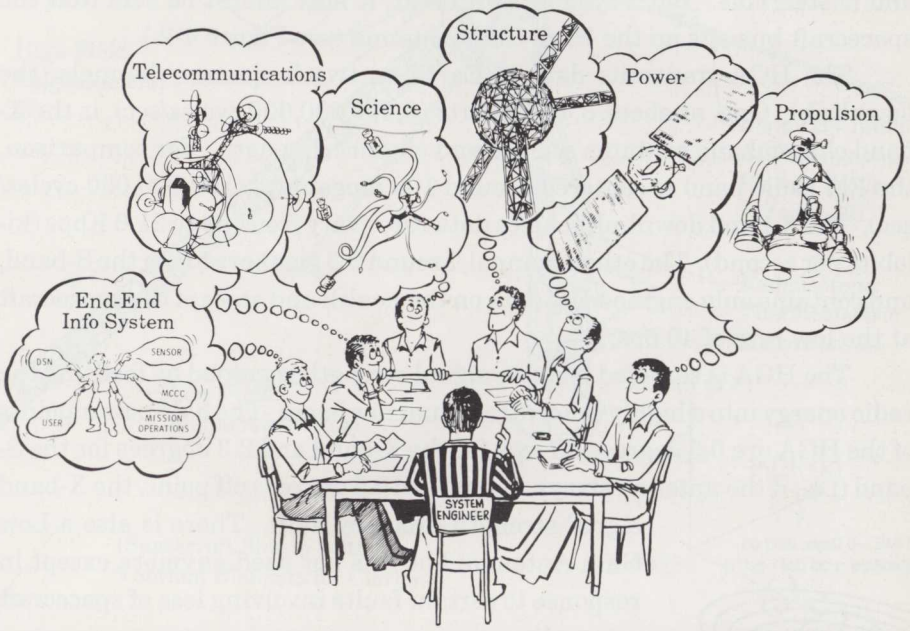


Figure 4-1. Before launch, a spacecraft design team did a lot of brainstorming to hammer out the dozens of major considerations (and thousands of smaller details) needed to design and build the amazing Voyager robots.

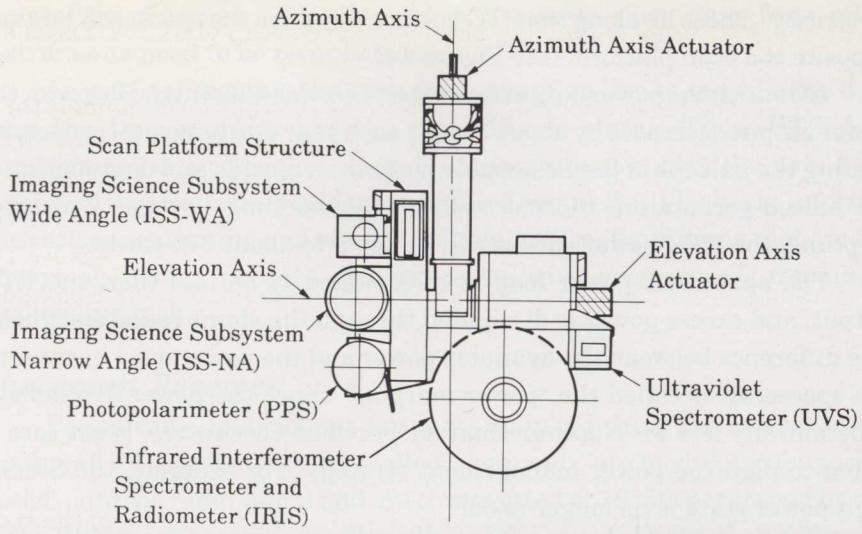


Figure 4-4. This view of the Voyager scan platform shows the locations of the two electric motors and gear trains, known as "actuators," that drive the platform to look in different directions.

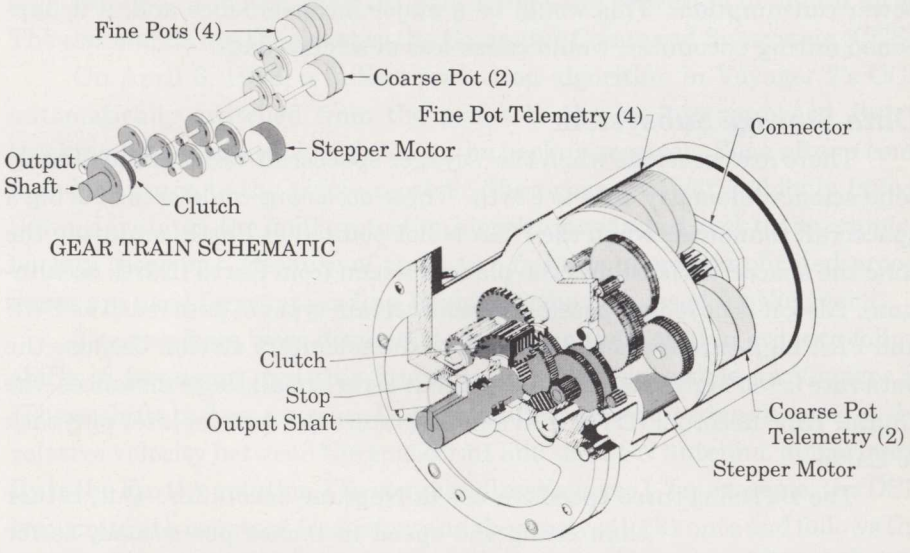


Figure 4-5. Small electric motors drive the Voyager scan platform about "azimuth" and "elevation" axes. Voyager 2's azimuth actuator stuck shortly after the Saturn encounter, but was used for the Uranus encounter and will be used for the Neptune encounter.

required for a response to the problem from Earth: over eight hours round-trip light time (at Neptune closest approach), plus the reaction time of the engineers to detect the problem and prepare the proper response. In many instances such a delay would be intolerable.

The other CCS function, storing and processing commands from Earth, allows the spacecraft to act as an intelligent robot to carry out its science-gathering functions in strict accordance with the carefully developed mission plan.

It is convenient to send up one transmission to the spacecraft which contains most or all of the commands needed to operate it for periods of time ranging from 30 days during the Observatory Phase to only two days during Near Encounter. Each of these transmissions nearly fills the remaining (non-FPA) memory in the two CCS computers. The contents of each of these transmissions is called a CCS Load and is given an identifying number. "A" loads refer to Voyager 1, and "B" loads to Voyager 2; for example, A821, B901, B902, etc. Chapter 6 contains a timeline which shows each of the encounter CCS Loads.

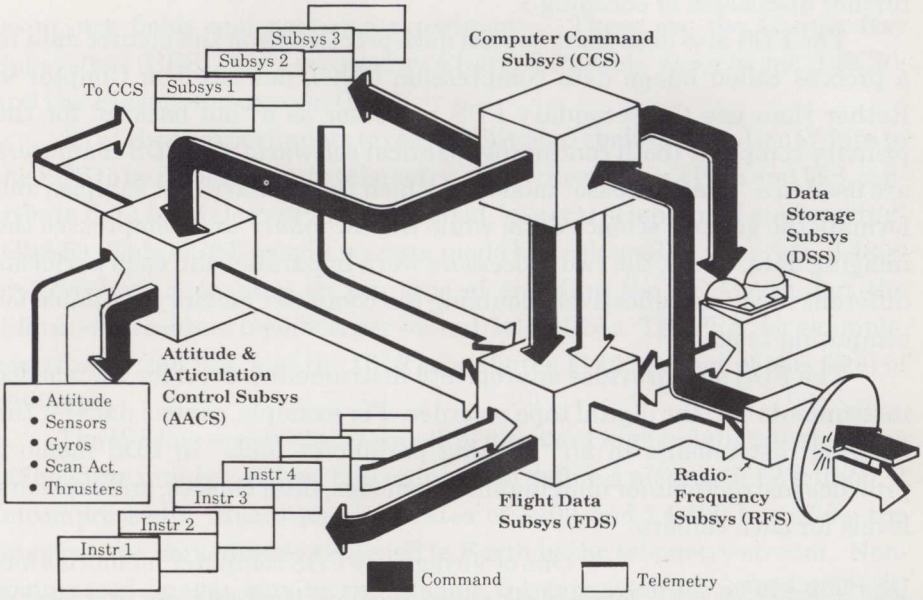


Figure 4-6. Voyager's three computer subsystems contain nearly 33,000 words of memory storage, with the Computer Command Subsystem (CCS) directing most of the activities.

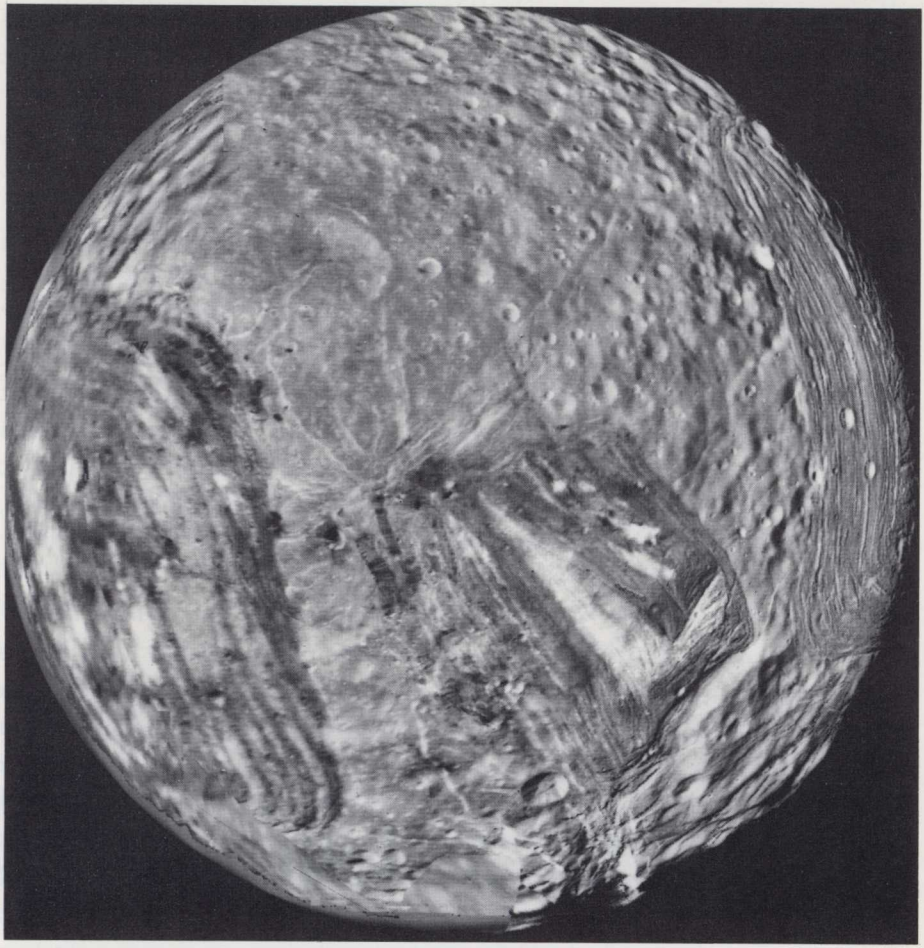


Figure 13-14. This image of Uranus' moon Miranda—the highest-resolution image ever returned by either Voyager spacecraft—shows the bizarre variety in the geology found there by Voyager 2. The smallest discernible features are less than 1 km (0.6 mi) across.

have faster periods of revolution (8 to 15 hr) than their more distant relatives. Puck, the tenth and largest new moon, and the first to be discovered by Voyager, is 154 km (96 mi) in diameter and makes a trip around Uranus every 18 hours.

★ Two of the moons discovered by Voyager, Cordelia and Ophelia, are shepherding satellites that gravitationally constrain the outermost ring, epsilon. The other eight new moons lie between the orbits of Ophelia and Miranda.